

Homework 5

Exercises

1. Problem #4 from Section 4.3 of *Mathematical Reasoning: Writing and Proof* [1].

Note: Henceforth, the course is designed to rely on material **that we have explicitly constructed**. Unless explicitly directed otherwise, you should not use theorems that you have learned in other courses, or other such concepts, unless we have verified them ourselves. You may only use results (Axioms and Propositions) from the last week in your proofs! Also recall that we use the convention $\mathbb{N} := \mathbb{Z}^+ \cup \{0\}$.

2. Let R be a commutative ring and $x, y, z \in R$. Prove

- (a) $(-x)(-y) = xy$.
- (b) $x(y - z) = xy - xz$.
- (c) $(-1)x = -x$.

3. Let R be an ordered ring and take $x, y, z \in R$ with $x < y$ and $0 < z$. Prove that $xz < yz$.

4. Henceforth, we define $\mathbb{Z}$ as the (unique) ring satisfying Axioms 1–12 in the Lecture Notes. We will derive all properties of $\mathbb{Z}$ purely from these Axioms! Arguing by contradiction, **prove there is no smallest or largest element of $\mathbb{Z}$** .

Reminder: You should prove this using only our Axioms and the ensuing Propositions! We don't have the Archimedean Principle yet—we don't even know what a real number is!

5. Use induction to show that $\sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$.

6. Use induction to show that, for any $r \neq 1$ and $n \in \mathbb{Z}^+$, we have $\sum_{k=0}^n r^k = \frac{r^{n+1} - 1}{r - 1}$.

7. Find and prove a formula for the sum $\sum_{k=1}^n (2k - 1)$ for all $n \in \mathbb{Z}^+$.

8. Find and prove a formula for the sum $\sum_{k=1}^n \frac{1}{k(k+1)}$ for all $n \in \mathbb{Z}^+$.

9. Recall that, for all $n \geq 2$, we define “ n factorial” (denoted $n!$) as

$$n! = n \cdot (n - 1) \cdots 2 \cdot 1.$$

We also set $0! = 1! = 1$. Argue by induction that $n! > 3^n$ for all $n \geq 7$.

References

- [1] Ted Sundstrom. *Mathematical Reasoning: Writing and Proof*. Grand Valley State University Libraries, 3rd edition, 2020.